

GOKULNAATH GOVINDARAJ

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SUMMARY

Backend and AI systems engineer with production experience building high-throughput Python APIs, async RabbitMQ-backed pipelines across 8 concurrent workers, and RAG retrieval systems at an NEC-backed startup. Diagnosed a ~2,000 RPS PostgreSQL bottleneck reducing p95 latency from 1.8s to 200ms and drove RAG recall to 93% across a 600-query evaluation framework and is authorized to work in the US on OPT through February 2029 with no sponsorship required.

HONORS & RECOGNITION

- Runner-Up, Wright Venture 2026 - AccioFirst** Wright State University · April 2026 \$2,000 awarded
- Technical Lead, NSF I-Corps Cohort** University of Connecticut · Sep 2024 \$1,000 awarded
- Owned the Spring Boot backend architecture for a peer-mentoring platform, coordinating API contracts with the front-end team and integrating an LLM-powered chatbot via Spring AI.

SKILLS

AI / LLM Systems: LLM API integration (Claude, Gemini 2.5, GPT-4.1), RAG pipelines, hybrid retrieval (BM25 + dense), Weaviate, OCR ingestion (Docling, PyMuPDF), prompt engineering, LLM evaluation frameworks, agentic orchestration (Claude Code CLI, n8n)

Languages: Python, Java (Spring Boot, Android), JavaScript / TypeScript, SQL, Go

Backend & APIs: FastAPI (async, Pydantic validation), Flask, RESTful API design, GraphQL, SQLAlchemy (ORM, Alembic), Spring Boot, RabbitMQ (async job queues, concurrent workers), microservices architecture

Infrastructure & Databases: PostgreSQL (schema design, indexing, connection pooling), Docker, GCP (Vertex AI, Cloud Run, GKE), AWS (RDS, S3, EC2), Kubernetes (kubebuilder/envtest), GitLab CI/CD, MongoDB (GridFS)

EXPERIENCE

- AI Trainer (Software Engineering & Agentic Workflows)** - Handshake AI Apr 2026 - Present | Remote - Contract
- Deliver golden-solution pull requests verified by Pass-to-Pass and Fail-to-Pass automated validation gates on HAI's Helix project, applying Claude Code CLI agents, Ralph loop autonomy patterns to generate elite training data for advanced AI models.
 - Author atomic prompt scaffolding and clear completion criteria that let headless AI agents iterate autonomously across long debugging sessions while preventing context pollution via Git and filesystem rotation.

- Software Engineer** - AriesView (NEC-backed) Aug 2025 - Present | Boston, MA
- Architected the RAG pipeline for a CRE due diligence platform, designing chunking strategies, embedding pipelines, and hybrid BM25 + dense retrieval against a Weaviate vector store to power document-aware Q&A over financial and lease documents.
 - Redesigned the RAG evaluation framework with 15 legal and financial documents and two 300-query sets after identifying that initial precision (0.1x) reflected insufficient test coverage rather than pipeline failure, establishing benchmarks 93% recall.
 - Diagnosed PostgreSQL connection pool saturation under ~2,000 RPS using k6 load tests, reducing p95 API latency from 1.8s to 200ms and cutting timeout errors from 1.2% to 0.1% by right-sizing Unicorn workers on an 8-vCPU deployment.
 - Orchestrated asynchronous OCR and LLM extraction across 8 concurrent workers via a RabbitMQ-backed job queue, enabling parallel multi-file document ingestion with metadata-driven job tracking and decoupled producer/consumer workflows.
 - Engineered a GraphQL pagination layer serving 100+ OCR-extracted document units to the React frontend one at a time on user navigation, replacing a bulk-fetch approach that would have overloaded the database and server.

- Programming Intern** - Porter Lee Corporation May - Aug 2024 | Schaumburg, IL
- Navigated a 30,000+ line Java Android codebase to build REST API endpoints for an evidence management system, reducing JSON deserialization time by ~180ms per request through optimized parsing logic.
 - Implemented JWT-based authentication securing API access across all endpoints and introduced multithreaded background sync that reduced average UI freeze duration by ~300ms during data refresh cycles.

- Programmer Analyst Trainee** - Cognizant Sep 2022 - Mar 2023 | Chennai, India
- Identified 10+ critical production defects in an unfamiliar Selenium and Java QA codebase during Agile release validation, escalating each to the development team before release.

PROJECTS

- RAG System - Financial Document Intelligence** Python · Weaviate · Hybrid BM25 + Dense Retrieval · LLM Evaluation
- Designed an end-to-end pipeline over financial PDFs using hybrid BM25 + dense retrieval against Weaviate, iterating across a 200-query evaluation framework until citation failures dropped 30% and retrieval recall reached 93%.

- AccioFirst - Job Intelligence Platform** FastAPI · Next.js · BrightData · Gemini 2.5 Flash-Lite · Supabase
- Built a multi-signal job intelligence scoring pipeline in FastAPI that ingests a job URL, orchestrates BrightData scraping, career-page cross-referencing, and company hiring-signal retrieval, then feeds structured output to Gemini 2.5 Flash-Lite to return a scored ghost-job verdict in under 5 seconds.

EDUCATION

M.S. Computer Science | Wright State University

Aug 2023 - Dec 2025